From Reconstruction to Anticipation

Predicting **future tissue geometry** from monocular laparoscopic video — with an honest estimate of its own confidence.

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What the project is for

Reconstruct a surgical scene in 3D from one laparoscope camera — then anticipate how the tissue deforms next.

Reconstruct
4D geometry from monocular video
COMPLETE · FYP-1

Learn deformation
Self-supervised, no labels
COMPLETE · MODULE 5

Predict ahead
1–3 frames, with confidence
IMPLEMENTED · MODULE 6

Evaluate
Against baselines and literature
IN PROGRESS · M7–M8

RESEARCH QUESTION

Can future tissue geometry be predicted from ordinary surgical video — and can the system say how much to trust each prediction?

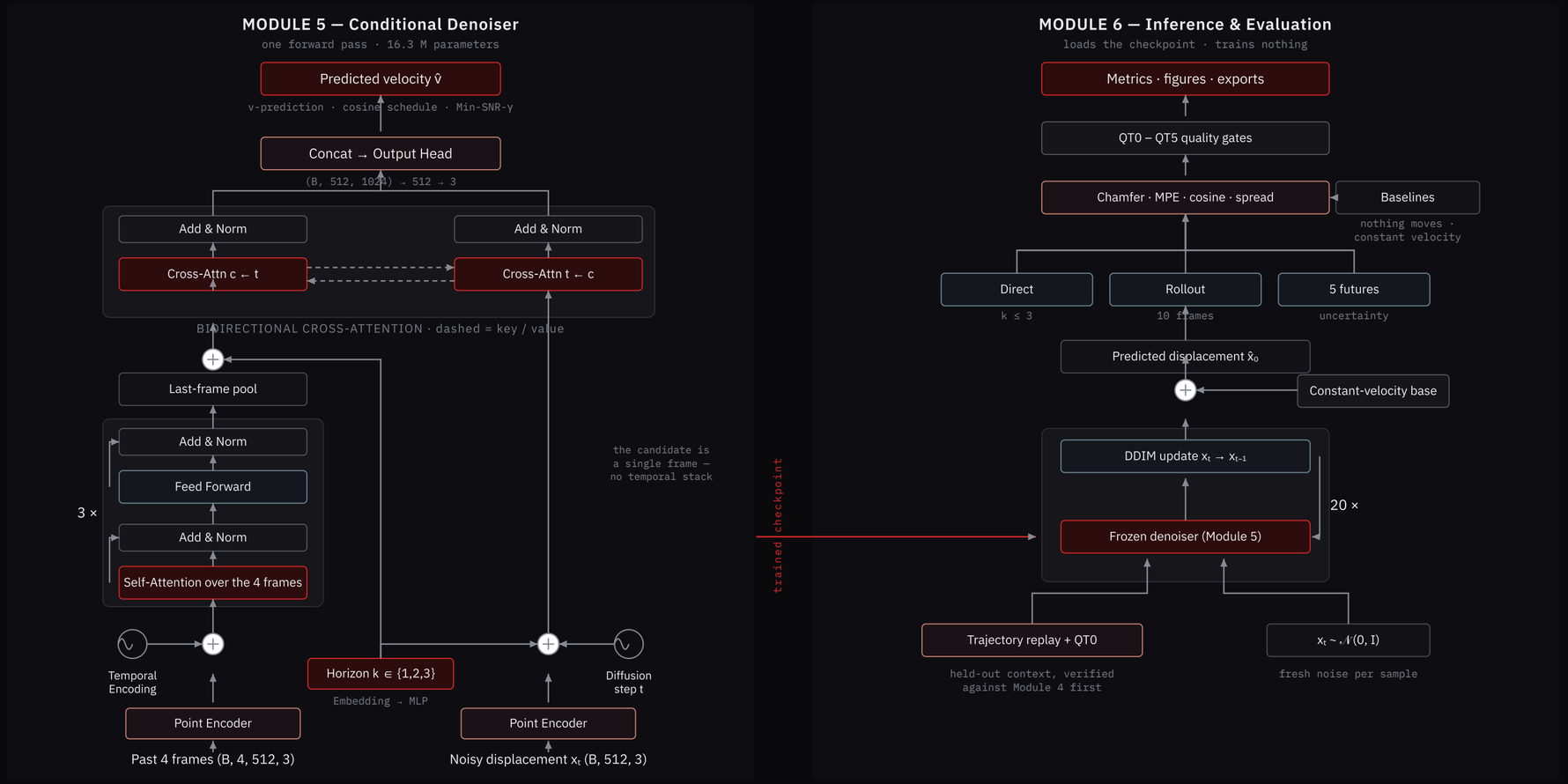
Faculty feedback from FYP-1, and what we did

Implement a base paper for comparative analysis	**Two base papers implemented** PointRNN (Fan & Yang, 2019) · Luo & Hu (CVPR 2021) `module7_baselines.py`
Select a few SOTA papers and implement them	**Eleven methods on one protocol** Classical: zero-motion, constant velocity, constant acceleration, linear fit, Kalman. Learned: MLP, GRU, LSTM, Transformer, PointLSTM. `781 lines`
Detailed analysis — tuning, optimisation	**Four tuning studies** Inference cost vs accuracy · context length · optimiser sensitivity · diffusion ablations `module8_hpo.py`
Start a paper in IEEE or Springer format	**IEEE manuscript drafted** Six pages, 22 references, FYP-1 and FYP-2 as one comparative study `paper/main.pdf`

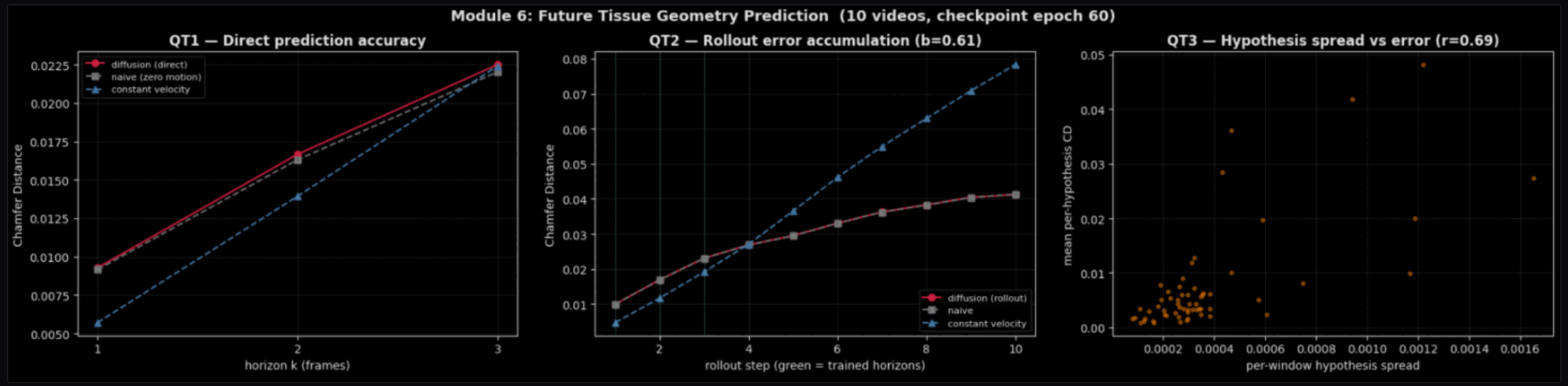
All four are implemented and in the repository. Two still need their compute runs — item 1 on the remaining-work slide.

Architecture — Modules 5 and 6

Given the last 4 frames of 512 tracked points, predict their displacement 1–3 frames ahead — trained on 10 videos, no clinical labels.



Metrics chosen, and what they measured



ACCURACY — NOT YET AHEAD

Constant velocity beats us at every horizon, on all ten videos.

DIRECTION — CLEARLY LEARNED

Cosine +0.57 at t+1. A model that learned nothing reads 0.00.

UNCERTAINTY — CALIBRATED

Spread across five futures tracks realised error, r = 0.69.

What we build on, and where we go past it

THE WORK WE BUILD ON	WHERE IT STOPS → WHAT WE ADD
EndoGaussian · Deform3DGS surgical reconstruction	**Only ever describe what already happened** We match them from one camera, then predict forward.
PointRNN point-cloud forecasting · our main learned baseline	**Rigid outdoor scenes, one answer, no confidence** Deforming soft tissue — and ours says how much to trust it.
Luo & Hu, CVPR 2021 the first point-cloud diffusion model	**Generates shapes; does not forecast** Ours is that idea made conditional on time.
TPG-VAE · SurgVista surgical anticipation	**Flat pixels, and told the instrument's action** We predict 3D geometry, and we are not told the action.

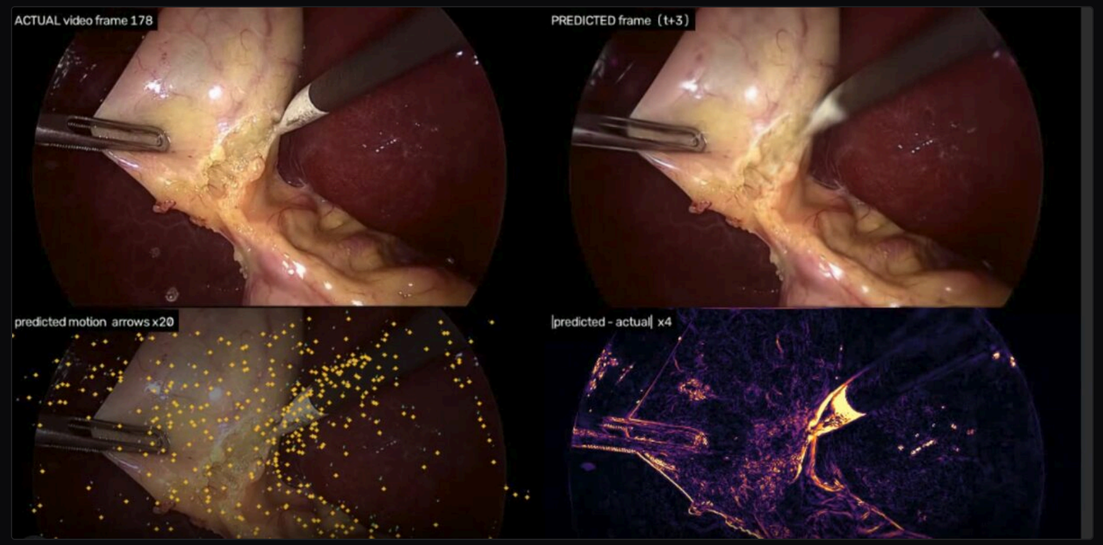
No published method does all of it at once — predict the ***future***, in 3D, on soft tissue, from a single camera, and say how confident it is.

Remaining work until the final evaluation

1. **Run the 11-method comparison and the HPO studies**
   Both written and tested. What remains is compute, not code.
2. **Evaluate the two corrections already implemented**
   Calibration needs no retraining; the residual target needs one retrain.
3. **Move to leave-one-video-out evaluation**
4. **Fold the finished results into the manuscript**
5. **Submit, and extend the corpus beyond 10 recordings**

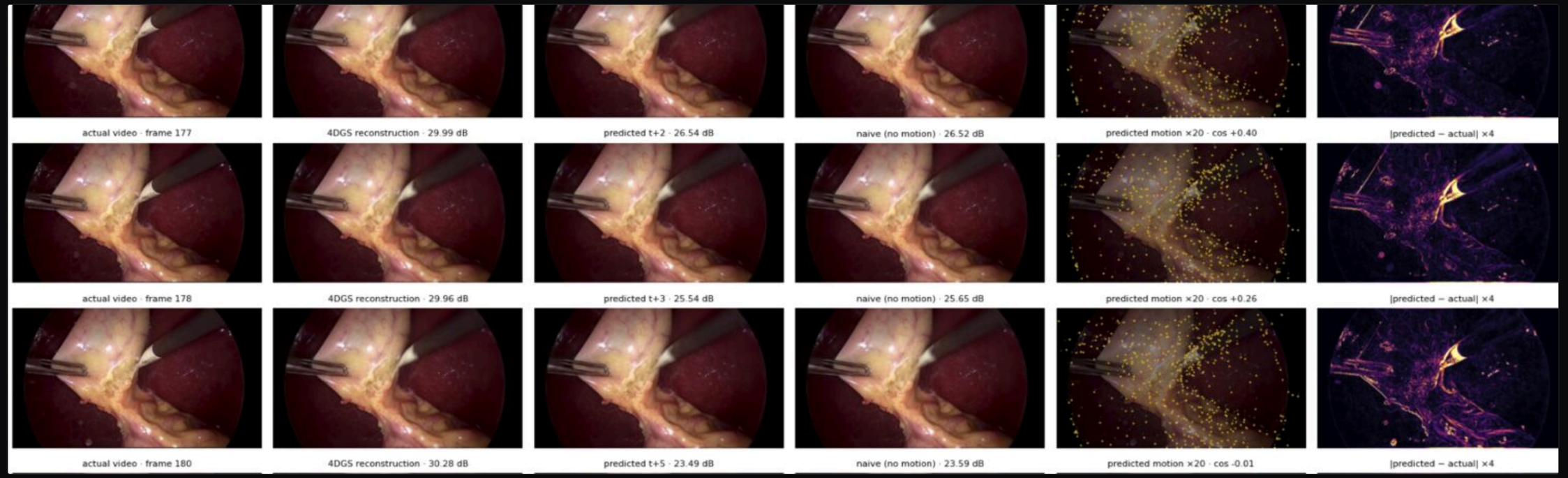
What you are about to see, running live

1. **Load the checkpoint, verify the data**
   QT0 gates everything downstream.
2. **Predict a held-out moment at t+1 … t+3**
   Against both baselines, same window.
3. **Render the prediction as a video frame**
   512 points carried onto 58,000 Gaussians.
4. **Compare side by side with the real frame**
   With magnified motion arrows.
5. **Forecast past the end of the recording**
   Twelve frames beyond the last image.



Nothing here is pre-rendered. The checkpoint is loaded and inference runs in front of you.

Making the prediction visible

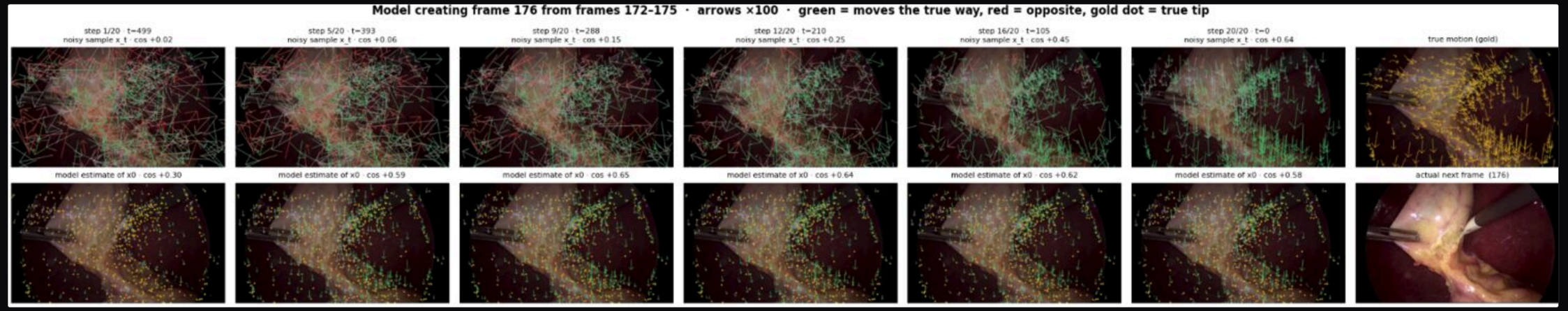


real frame · 4DGS reconstruction · our prediction · naive baseline · predicted motion ×20 · error ×4

Tissue moves one to two pixels a frame, so the arrows are magnified — the factor is printed on every panel and never applied to a reported number. The forecast runs twelve frames past the end of the recording.

30.04	26.79	26.74
RECONSTRUCTION CEILING, DB	OUR PREDICTION, DB	NAIVE BASELINE, DB

The model knows which way tissue moves — not how far



twenty denoising steps · green arrows move the true way, red oppose, gold is truth

t+1	t+2	t+3	t+5	t+10
+0.57	+0.40	+0.26	−0.01	−0.49

A model that learned nothing would read 0.00 everywhere. The direction is right; the *distance* is wrong — which is a far more tractable problem, and both fixes are implemented.

The model knows when it should not be trusted

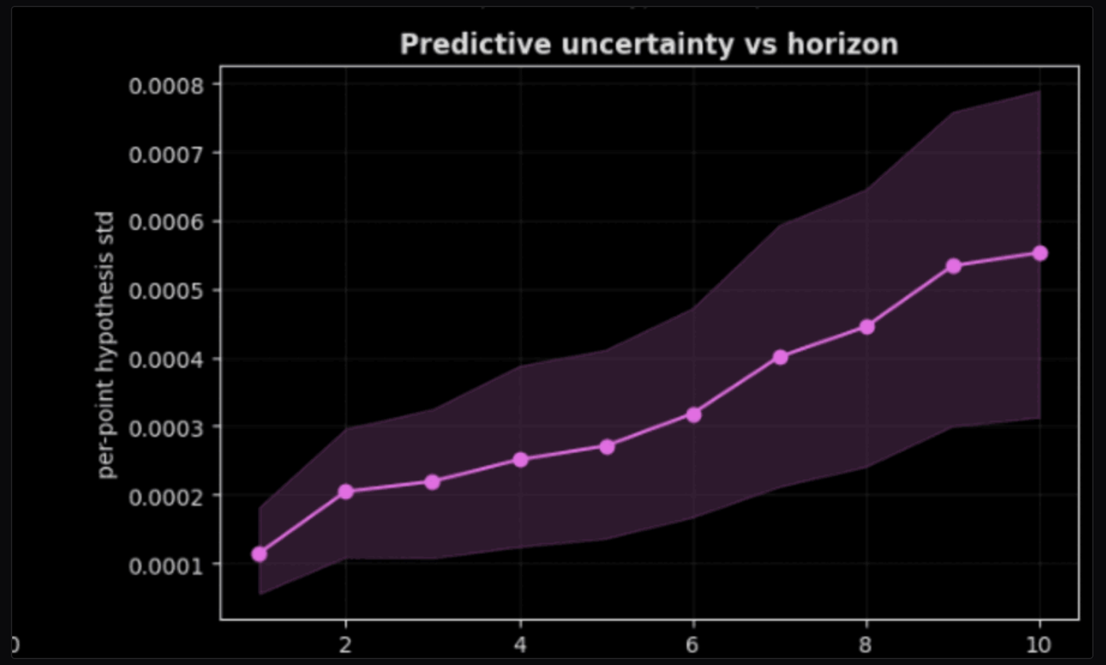
r = 0.69
SPREAD VS REALISED ERROR

5
FUTURES SAMPLED PER INPUT

5×
UNCERTAINTY GROWTH OVER 10 FRAMES

When the five sampled futures disagree, the prediction really is worse — so the system can flag its own weak moments. A model that is confidently wrong is unusable in theatre; one that says *“do not trust me here”* is usable even when imperfect.

Stated carefully: this is correlation between predicted spread and realised error. Some of it may reflect that busy moments are both harder and more variable.

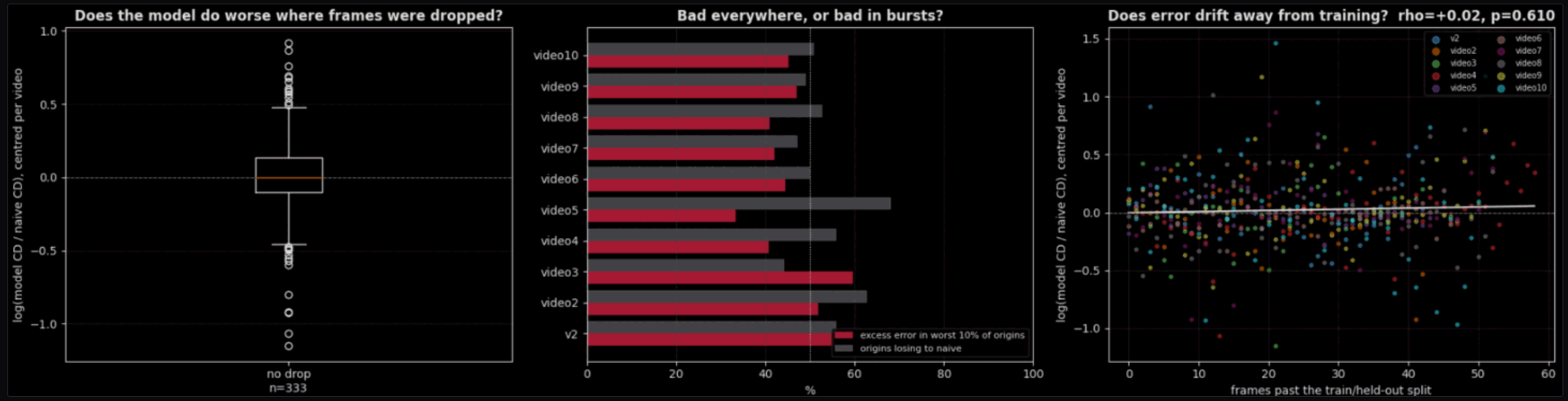


A two-line formula beats our model

METHOD	K = 1	K = 2	K = 3
Nothing moves (naive)	0.0093	0.0165	0.0222
Constant velocity	**0.0055**	**0.0138**	**0.0187**
Ours — diffusion	**0.0093**	**0.0165**	**0.0225**

Chamfer Distance, lower is better · full held-out set, 10 videos

We lose on all ten videos, so it is not a fluke we can explain away. Before concluding anything, we tested and eliminated the two obvious alternatives — and the diagnostics below are how.



no drift (ρ = +0.02, p = 0.61) · no dropped frames (333 origins checked, 0 found) · excess error is not concentrated in bursts